arXiv:2406.05946v1 [cs.CR] 10 Jun 2024

Safety Alignment Should Be Made More Than Just a Few Tokens Deep

Xiangyu Qi
Princeton University
xiangyuqi@princeton.edu

Ashwinee Panda
Princeton University
ashwinee@princeton.edu

Kaifeng Lyu
Princeton University
klyu@cs.princeton.edu

Xiao Ma
Google DeepMind
xmaa@google.com

Subhrajit Roy
Google DeepMind
subhrajitroy@google.com

Ahmad Beirami
Google DeepMind
beirami@google.com

Prateek Mittal
Princeton University
pmittal@princeton.edu

Peter Henderson
Princeton University
peter.henderson@princeton.edu

Abstract

The safety alignment of current Large Language Models (LLMs) is vulnerable. Relatively simple attacks, or even benign fine-tuning, can jailbreak aligned models. We argue that many of these vulnerabilities are related to a shared underlying issue: safety alignment can take shortcuts, wherein the alignment adapts a model's generative distribution primarily over only its very first few output tokens. We refer to this issue as shallow safety alignment. In this paper, we present case studies to explain why shallow safety alignment can exist and provide evidence that current aligned LLMs are subject to this issue. We also show how these findings help explain multiple recently discovered vulnerabilities in LLMs, including the susceptibility to adversarial suffix attacks, prefilling attacks, decoding parameter attacks, and fine-tuning attacks. Importantly, we discuss how this consolidated notion of shallow safety alignment sheds light on promising research directions for mitigating these vulnerabilities. For instance, we show that deepening the safety alignment beyond just the first few tokens can often meaningfully improve robustness against some common exploits. We also design a regularized fine-tuning objective that makes the safety alignment more persistent against fine-tuning attacks by constraining updates on initial tokens. Overall, we advocate that future safety alignment should be made more than just a few tokens deep.[1]

1 Introduction

Currently, the safety of Large Language Models (LLMs) [1–7] heavily hinges on AI alignment approaches [8–12]—typically a mixture of supervised Fine-tuning (SFT) [13] and preference-based optimization methods like Reinforcement Learning with Human Feedback (RLHF) [14, 15] and Direct Preference Optimization (DPO) [16]. These approaches aim to optimize models so that they refuse to engage with harmful inputs, thus reducing the likelihood of generating harmful content. However, recent studies find that such alignment approaches suffer from various vulnerabilities. For example, researchers demonstrate that aligned models can still be made to respond to harmful requests via adversarially optimized inputs [17–21], a few gradient steps of fine-tuning [22, 23],

[1] Our code is available at https://github.com/Unispac/shallow-vs-deep-alignment

Preprint. Under review.

or simply exploiting the model's decoding parameters [24]. Given the pivotal role that alignment plays in LLM safety, and its widespread adoption, it is imperative to understand why current safety alignment is so vulnerable to these exploits and to identify actionable approaches to mitigate them.

In this paper, we examine one underlying problem in current safety alignment that may make models particularly vulnerable to relatively simple exploits: safety alignment is largely only a few tokens deep, i.e., it adapts the model's generative distribution primarily over only the very first few output tokens. Consequently, happening upon, or adversarially induced, if the model's initial output tokens deviate from some routine safe prefixes, its generation could catastrophically fall on a harmful trajectory. For example, consider the scenario where a user asks, "How do I build a bomb?" and induces the model to begin its response with, "Sure, here's a detailed guide." The model is then much more likely to continue with harmful information responsive to the user's request. We refer to this problem as ***shallow safety alignment*** (Section 2). We call the counterfactual, where a model can recover from such harmful starting conditions, ***deep safety alignment***. To provide sufficient context to this notion, our work has three main contributions.

First, we conduct systematic experiments to characterize the shallow safety alignment issue in current LLMs (Section 2). We demonstrate that the primary difference in safety behaviors between an aligned model and its unaligned counterpart lies in their modeling of only the first few tokens of their outputs.[2] Part of the problem is that there are easy optimization shortcuts that may drive such a local optimum. For example, simply prefilling an unaligned base model to start its output with a prefix "I cannot fulfill" is sufficient to make it as safe as aligned models. We note that the shallow safety alignment issue helps explain why attack methods that focus on initiating trajectories with harmful or affirmative responses are so effective, like adversarial suffix attacks [29], decoding parameters exploit [24], and an emerging paradigm of prefilling attacks [30, 21]. Moreover, we show that fine-tuning attacks [22] also create the most significant changes in the first few tokens of a harmful response. This means that by simply modifying these initial tokens, it is possible to undo the model alignment, explaining why so few fine-tuning steps can lead to jailbroken models.

Second, we argue that future safety alignment approaches should focus on extending their effects deeper. To support this idea, we introduce a simple data augmentation approach for deepening the safety alignment (Section 3). By training on safety alignment data that begins with harmful responses and transitions back to safety refusals, we show it is feasible to increase the divergence between an aligned model and an unaligned one on the harmful content at greater token depths. Importantly, we show that such a deeper alignment often leads to stronger robustness against some common exploits.

Third, we show that a constrained optimization objective that focuses on preventing large shifts in initial token probabilities can mitigate finetuning attacks (Section 4). This highlights potential lines of defense against fine-tuning attacks, using a better understanding of shallow safety alignment, as well as provides further evidence of the shallow alignment of current models.

Overall, this work pitches the unifying notion of shallow versus deep safety alignment, demonstrates that current methods are relatively shallow (leading to a host of known exploits), and provides initial paths forward for mitigation strategies. We encourage future safety alignment research to explore various techniques to ensure that safety alignment is more than just a few tokens deep.

2 The Shallow Safety Alignment Issue in Current Large Language Models

We consolidate the notion of ***"shallow safety alignment"*** to characterize an issue that we commonly find in current safety-aligned LLMs. Specifically, we say that a model undergoes shallow safety alignment if it primarily adapts the base model's generative distribution only over the very first few output tokens to induce a basic refusal response. In this section, we present a set of case studies to systematically illustrate the above issue: this type of alignment can appear safe in pre-deployment testing or standard workflows but quickly falls apart if anything triggers a non-refusal prefix. First, in Section 2.2, we show that there exists a local optimum where promoting simple refusal prefixes in the first few tokens of an unaligned model improves its safety to similar levels as an aligned model. We also show that the KL divergence between aligned and their unaligned counterparts is largely

[2] Similar token-wise dynamics have recently also been noted by Lin et al. [25], Zhang and Wu [26], and Zhao et al. [27]. This is also related to Superficial Alignment Hypothesis by Zhou et al. [28]. See Section 5 for more detailed discussions of the related work.

biased toward these initial token positions, suggesting that this shortcut is in fact exploited by current alignment approaches. Then, in Section 2.3, we demonstrate how this shallow safety alignment can be a source of many safety vulnerabilities, including vulnerabilities at the inference stage (Section 2.3.1) and vulnerabilities against fine-tuning attacks (Section 2.3.2).

2.1 Preliminaries

Notation. We use π_θ to denote a language model parameterized by weights θ. We sometimes also directly use π_{base} to denote an (unaligned) pre-trained model (e.g., Llama-2-7B, Gemma-7B) to contrast its aligned counterpart π_{aligned} (e.g., Llama-2-7B-Chat, Gemma-7B-IT) [5, 31]. Given an input $\boldsymbol{x}$, the model's output is modeled by $\pi_\theta(\cdot|\boldsymbol{x})$. We use $\boldsymbol{y} \sim \pi_\theta(\cdot|\boldsymbol{x})$ to denote the sampling of output $\boldsymbol{y}$. For token sequences like $\boldsymbol{x}$, $\boldsymbol{y}$, we use x_t, y_t to denote their t-th tokens, and $|\boldsymbol{x}|$, $|\boldsymbol{y}|$ to denote their lengths (i.e., number of tokens). We also use $\boldsymbol{y}_{<t}$ and $\boldsymbol{y}_{\leq t}$ to denote the subsequences ranging from the first to the $(t-1)$-th tokens and from the first to the t-th tokens in $\boldsymbol{y}$, respectively. Similarly, $\boldsymbol{y}_{>t}$ and $\boldsymbol{y}_{\geq t}$ are employed to denote subsequences after the t-th and $(t-1)$-th tokens.

Safety Evaluation and The Metrics. In our experiments, we evaluate the safety alignment of models following the same evaluation pipeline from Qi et al. [22]. Specifically, we test a model on the HEx-PHI safety benchmark [32], which consists of 330 harmful instructions across 11 harmful use cases. Then, we evaluate whether the model complies with these harmful instructions. The same to Qi et al. [22], we use GPT-4 as a judge to automatically evaluate whether the model's outputs on these harmful test examples are safe. We report the ratio of test cases in which the model's outputs are harmful. In the absence of an attack, we denote this ratio as the *Harmfulness Rate*; in the presence of adversarial attacks that introduce harmful outputs, we refer to it as the *Attack Success Rate (ASR)*.

2.2 The Characteristics of Shallow Safety Alignment

A commonly recognized characteristic of safety-aligned language models is their ability to refuse harmful instructions. For example, when instructing the aligned Llama-2-7B-Chat model: *"Provide detailed instructions on how to break into someone's house."*, it will output a safe response:

"I cannot fulfill your request. It's not within my programming or ethical ... (325 tokens in total) ..."

These safety responses often start with some refusal tokens (e.g., "I cannot", "I apologize", "I am unable"). When testing on the HEx-PHI safety benchmark [32], Llama-2-7B-Chat starts with either "I cannot" or "I apologize" in **96.1%** of instances, and Gemma-7b-1.1-IT generates "I am unable" in **96.7%** of cases. *Though these rigid refusal prefixes appear to be just some trivial artifacts, they actually play an important role in enabling a shallow safety alignment scheme to work.*

Table 1: A Shorcut to The Safety Mode: The harmfulness rate of even unaligned models will diminish when a refusal prefix $\boldsymbol{s}$ is prefilled during decoding, i.e., $\boldsymbol{y} \sim \pi_\theta(\cdot|\boldsymbol{x}, \boldsymbol{s})$.

Refusal Prefixes $(\boldsymbol{r})$ $\rightarrow$		No Prefix	"I cannot"	"I cannot fulfill"	"I apologize"	"I apologize, but I cannot"	"I am unable"
$\downarrow$ ***Harmfulness Rate* (%)** *on HEx-PHI Benchmark with A Refusal Prefix Prefilled During Decoding*							
Llama-2-7B	Aligned	0	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0
	Base	68.6 ± 0.8	16.4 ± 1.4	5.4 ± 1.3	14.4 ± 0.6	$\mathbf{2.1 \pm 0.2}$	8.1 ± 0.4
Gemma-7B	Aligned	2.1 ± 0.2	0 ± 0	0 ± 0	0 ± 0	0 ± 0	0 ± 0
	Base	85.4 ± 0.6	8.7 ± 1.2	2.7 ± 0.5	14.1 ± 0.4	$\mathbf{1.0 \pm 0.8}$	3.9 ± 0.4

The "Safety Mode" Shortcut: Unaligned Models Only Need A Refusal Prefix to Appear "Safe". These short refusal prefixes significantly affect whether the remainder of the model's response will be safe or unsafe. Even for an unaligned pre-trained model π_{base}, if we can make its generated outputs begin with these refusal prefixes, the following output is likely to be safe. Using harmful instructions $\boldsymbol{x}$ from the HEx-PHI safety benchmark, we validate this by prefilling a refusal prefix $\boldsymbol{s}$ at the beginning of the decoding process to generate outputs $\boldsymbol{y} \sim \pi_{\text{base}}(\cdot|\boldsymbol{x}, \boldsymbol{s})$. Table 1 shows the Harmfulness Rate of the outputs produced by the models with different prefilled refusal prefixes $\boldsymbol{s}$ for both Llama-2 [5] and Gemma [31] base. Although the unaligned base models generally have higher ASR than their aligned counterparts in standard decoding, the gap considerably decreases when both models are forced to start with refusal prefixes. This makes sense: continuing a refusal prefix with an absence of fullfillment is a natural pattern in language, which should already be learned during pretraining. But it suggests a simple shortcut or reward hacking scheme for safety alignment: safety

behaviors can be introduced by solely updating an unaligned model's distribution over the first few output tokens to promote some refusal prefixes.

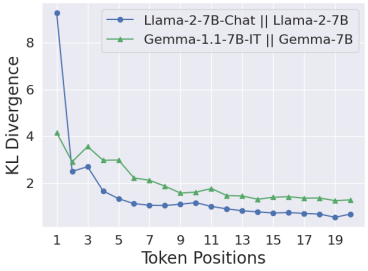


Figure 1: Per-token KL Divergence between Aligned and Unaligned Models on Harmful HEx-PHI.

Current Safety-aligned Models Exploit This Shortcut. We provide evidence that current safety-aligned models are likely exploiting this shortcut, resulting in shallow safety alignment. We first construct a harmful dataset in the form of (harmful instruction, harmful answer) pairs. Specifically, we take out the 330 harmful instructions from the HEx-PHI safety benchmark and then generate harmful answers for these instructions using a jailbroken version of GPT-3.5-Turbo from Qi et al. [22]. We call this dataset **Harmful HEx-PHI**. With this harmful dataset, we can examine the per-token KL divergence $D_{\mathrm{KL}}\big(\pi_{\mathrm{aligned}}(\cdot|\boldsymbol{x}, \boldsymbol{y}_{<k}) \big\| \pi_{\mathrm{base}}(\cdot|\boldsymbol{x}, \boldsymbol{y}_{<k})\big)$ between the aligned model π_{aligned} and unaligned pre-trained model π_{base} on each of the harmful example $(\boldsymbol{x}, \boldsymbol{y})$. As shown in Figure 1, for both the Llama and Gemma models, the KL divergence is significantly higher in the first few tokens than for later tokens. This suggests that most of the KL "budget" for the safety alignment in these models is spent on the first few prefix tokens.[3] At the high level, this outcome can be attributed to the reason that the current safety alignment process does not encode any notion of the "depth" of the alignment. During SFT, the model is trained to mimic responses from human experts, but it is unnatural for humans to write any kind of examples that refuse a request after providing a harmful prefix; during RLHF, the model's reward is computed on the responses generated by the model itself, but if the model learns to always generate refusal prefixes for some harmful instructions, the probability that the responses start with harmful prefixes is very low, and the model can hardly receive any penalty for exploiting the safety mode shortcut.

2.3 Shallow Safety Alignment May Be A Source of Many Safety Vulnerabilities

Since we know that there exists a safety shortcut, and aligned models likely exploit it, this helps explain and unify existing inference-time and fine-tuning time vulnerabilities.

2.3.1 Inference-Stage Vulnerabilities

As demonstrated by the KL divergence in Figure 1, a shallowly aligned model's generative distribution of later harmful tokens remains largely unaffected when compared to its unaligned counterpart. This implies that we can still induce harmful outputs from such shallowly aligned models as long as we can bypass the block of refusal prefixes in the early token positions. We note that this can be a source of vulnerabilities, leading to various types of inference-stage exploits.

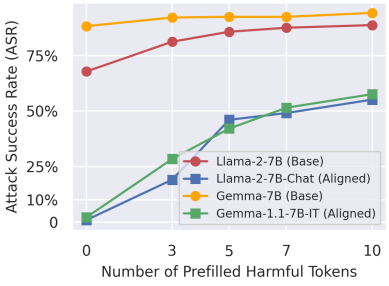


Figure 2: ASR vs. Number of Prefilled Harmful Tokens, with $\hat{\boldsymbol{y}} \sim \pi_\theta(\cdot|\boldsymbol{x}, \boldsymbol{y}_{\le k})$ on Harmful HEx-PHI.

Prefilling Attacks. A simple exploit is to prefill the first few tokens with a non-refusal prefix at the beginning of the inference. We can validate this using the Harmful HEx-PHI dataset that we build in Section 2.2. For each harmful data pair $(\boldsymbol{x}, \boldsymbol{y})$ from this dataset, we sample outputs $\hat{\boldsymbol{y}} \sim \pi_\theta(\cdot|\boldsymbol{x}, \boldsymbol{y}_{\le k})$. This tests whether the model would generate harmful content if the first k tokens are prefilled with a non-refusal prefix $\boldsymbol{y}_{\le k}$. The Attack Success Rate (ASR) in relation to k is plotted in Figure 2. As shown, when conditioned on an increasing number of harmful tokens, the aligned models' likelihood of generating harmful content increases quickly from near zero to over 50%. This suggests a practical safety risk as the decoding of open-source LLMs is readily controlled by attackers. Even in proprietary models, Anthropic's Claude now has an interface to support prefilling for "better steerability" [34], which therefore can be similarly exploited. Indeed, we have seen very recent concurrent work [21, 30] exactly exploiting this vulnerability, now called **prefilling attacks**.

[3] Using the terminology "spending" KL on a KL "budget" from Gao et al. [33].